

9.3 Direct Preference Optimization

The same pairwise data, no separate reward model and no PPO loop.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Direct Preference Optimization (DPO) (Rafailov et al., 2023) fits a policy to chosen/rejected pairs **without** training an RM and **without** PPO. The reward is written in closed form from the policy and a frozen reference, then the Bradley–Terry model becomes a classification loss on logits.

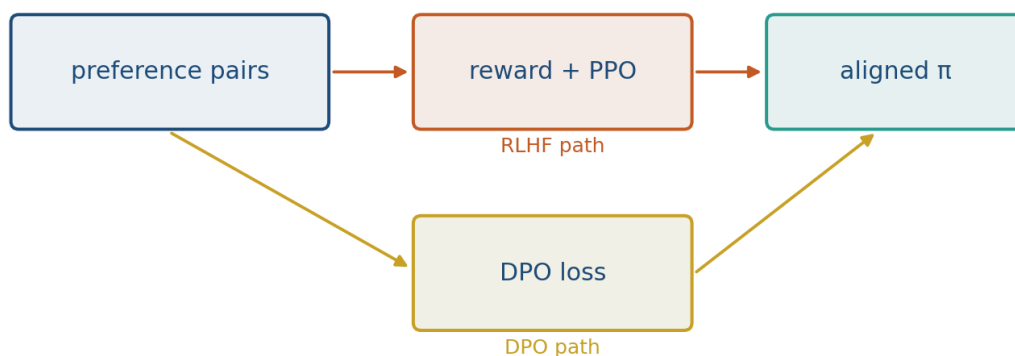
1. Skip the reward model

RLHF solves $\max_{\pi} \mathbb{E}[r] - \beta \text{KL}(\pi \| \pi_{\text{ref}})$. Under Bradley–Terry, the optimal r can be expressed as

$$r(x, y) = \beta \log \frac{\pi(y | x)}{\pi_{\text{ref}}(y | x)} + \text{const}(x).$$

Plug that into the RM logistic loss and the r parameters disappear. You train π_{θ} directly.

DPO fits the same pairwise data without an explicit reward model



2. The loss

$$\mathcal{L}_{\text{DPO}} = -\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_{\theta}(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right).$$

In a batch you need log-probabilities of **both** completions under π_{θ} and under π_{ref} . The reference stays frozen (usually SFT). β is the same kind of leash as in RLHF: it scales how hard you push away from π_{ref} .

Lab 9 will compute this on toy scalar logits, not on an LLM.

3. When DPO is the right picture

You have pairs, a GPU, and no appetite for a PPO stack. Offline: the pair set is fixed, so you do not explore new y unless you collect more. If the SFT model never proposed good y_w , DPO cannot invent them from nothing. Implicit reward can still overfit the pair set.

Report β , the reference checkpoint, and a held-out preference accuracy (how often π_θ assigns higher log-prob to y_w than to y_l). That number is not "the model is aligned"; it is "the model copied this rater."

4. Practice

1. If $\pi_\theta = \pi_{\text{ref}}$, what is $\mathcal{L}_{\text{DPO}}$? (Compute $\sigma(0)$.)
2. Why do you still keep a frozen reference if there is no RM?
3. Name one thing PPO can do that offline DPO cannot, given a fixed pair file.